EVIDENCE OF POLAR SHIFT SINCE TRIASSIC TIME

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ABSTRACT

The magnetizations of 343 samples of Permian and Triassic sediments from the United States are compared to magnetizations that have been reported for sediments of approximately the corresponding age in England. It is demonstrated that an impressive number of the observations are in essential agreement simply on the basis of assuming that the rocks were magnetized by a geomagnetic field about like the one today, the essential difference being that this field was in a significantly different orientation. This result is discussed in terms of slip of an outer shell of the earth relative to the axis of revolution.

INTRODUCTION

Since the appearance of the paper by Johnson, Murphy, and Torreson [see 1 of "References" at end of paper] on the magnetizations of Pleistocene varved clays, there has been a basis for hoping that the history of the changes of the earth's magnetic field can be traced far back in time by studies in sedimentary rocks. In the past few years, a number of pertinent findings have been reported. Among the most interesting, given first *in extenso* by Clegg, Almond, and Stubbs [2], and confirmed by Creer, Irving, and Runcorn [3], is the fact that there is a regional consistency in the magnetizations of the late Paleozoic and Triassic sediments of England. These magnetizations were found to be in such a pattern that two obvious interpretations could be advanced, as follows: (1) In Triassic time, England was in its present location but the earth's magnetic field was so oriented that the dipole axis emerged in the vicinity of Kamchatka and southeast of Brazil; or (2) England since Triassic time has experienced a clockwise rotation on a vertical axis of some 34°. Also of interest is the evidence presented to show that the sense of the earth's field reversed during the span of time represented by the sediments.

Earlier studies of Paleozoic sediments in the eastern United States did not reveal as simple a pattern of magnetization [4, 5, 6]. It was established that within the Silurian of the folded Appalachian belt there are zones of sediments which are magnetized as though they had originally been laid down in South Africa [7]. Efforts to confirm this interesting situation by studies of sediments outside the geosyncline were futile [8], and it became obvious that in these rocks there was not the necessary internal consistency to permit an accurate determination of the former orientation of the earth's magnetic field.

The fact noted by Clegg, *et al.*, that the English sediments with the interesting magnetizations had at one time been buried to a depth estimated as 10,000 feet, suggested a parallel with the findings in the Appalachians. There the Silurian sediments, too, had at one time been deeply buried; in addition, however, they had been considerably deformed, and some of the discrepancies in the patterns of magnetization were attributed to deformation. Sediments with magnetizations either parallel to the earth's present field or greatly scattered were found commonly only in sections which had never been deeply buried.

It follows that the deductions made from observations in the once deeply buried sediments of England should be examined by studies elsewhere in contemporaneous beds which also had been subjected to elevated pressures and temperatures. These are abundantly available in the southwestern United States. It was already known that the magnetization directions in a flat-lying section of Permian sediments (in Ohio), which have always been lying at shallow depths, were greatly scattered [6]; see Figure 1. This paper is a first report of studies that are being made to

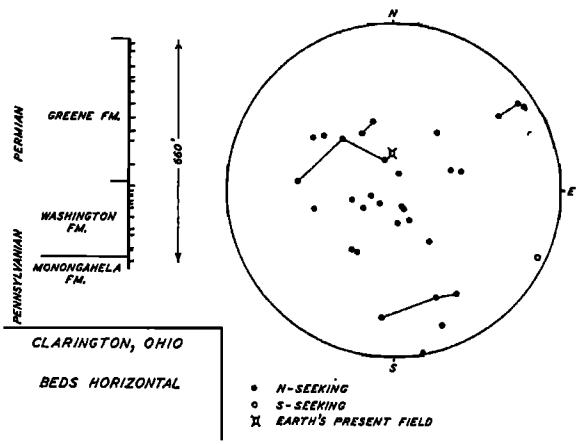


FIG. 1—Magnetizations of Permian sediments, Clarington, Ohio

determine the extent to which the findings in England can be confirmed and generalized by field observations of sediments of the same ages in the United States and elsewhere.

The criteria for selection of sites for sampling were mainly the following: Approximately flat-lying, unmetamorphosed, fine-grained sediments of sufficient coherence to permit preparation of oriented cores; reasonable accessibility; known position in the stratigraphic sequence. All samples which could be measured are reported; no localities were revisited for substantiating "interesting" results. In all, 301 oriented specimens from 12 different sites were collected and measured during the period March 22 to May 20, 1955.

ACKNOWLEDGMENTS

Dr. Charles B. Read, of the Fuels Branch, United States Geological Survey, at Albuquerque, New Mexico, and his colleagues, were most helpful in providing

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detailed information on a great many of the sites sampled, and in offering valuable discussion of some of the results as they were obtained in the field. Mr. W. N. Dove, of the Department of Terrestrial Magnetism, assisted during the first two weeks of the trip; his place was then filled by Mr. Kenneth G. Books, of the Geophysics Branch, United States Geological Survey, for two weeks; and he in turn was followed by Mr. William E. Huff, Jr., of the Geophysics Branch, USGS for the remainder of the trip. Coring and measuring oriented samples and driving a truck are arduous tasks, and I am very grateful for the cheerful and able assistance that was given. Finally, I wish to express my appreciation to Dr. E. J. Workman and Prof. Victor Vacquier, of the New Mexico Institute of Mining and Technology, for their cordial hospitality and help.

EXPERIMENTAL PROCEDURE

During the past eight years at the Department of Terrestrial Magnetism, a number of special procedures have been developed and used primarily for studies of the magnetism of sedimentary rocks [6, 9, 10, 11]. Since these have been of great help in the work, a detailed description of the important ones is given below.

Field truck—The truck is a Dodge Power Wagon (9,000 pounds gross weight), having available four-wheel drive and a winch on the front end. It is fitted with an aluminum van, 8 feet long, 6-1/4 feet wide, and 6-1/4 feet high, specially designed to contain all the equipment needed for collecting oriented core samples in the field and completing the measurements of their residual magnetism. The principal components are as follows:

For sampling—A 3.5-kva, 115-volt, ac generator for powering the drilling rig (more than adequate); a drilling rig which can cut cores at the outcrop to a depth of 3 inches; 55-gallon drum of water for cooling the drill-bit; water-pump, hoses, and extension cables which allow the drill-rig to be operated up to 100 feet from the truck; a standard drill-press which can be used when the drill-rig is impracticable; saws for end-cutting the cores to length.

For measuring—Spinner magnetometer (see below); devices for orienting cores and plotting final results.

Miscellaneous hand-tools, spare parts, camping equipment, bedding, etc. The design layout is successful in that it is possible to shift rapidly from one type of operation to another with a minimum of effort.

For the assistance of anyone contemplating constructing a similar field laboratory, the following criticisms of the systems now in use may prove valuable.

The drilling rig was fabricated from a small standard drill-press, a six-foot aluminum ladder, a 1/3-hp, 1725-rpm, ac motor, a flexible cable, and minor parts. It has proved exceedingly effective for sampling some exposures, and completely worthless in others. Its chief drawback is that it is too unwieldy for two men to handle on rough terrain. Occasionally some rocks are too tough to core effectively with the bit pressures that can be exerted on the standard bits which cut a one-inch core standing in a 1-1/8 inch hole. The self-centering behavior of a coring cutter minimizes the problem of accurate spindle alignment. The chief problem in using the drilling-rig is to assure at the start that the legs of the rig will remain

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stationary during the coring operation. Once the core has been started, care must be taken to maintain constant pressure on the bit so that its angle of attack is not changed as the frame of the rig is sprung by the applied pressure. This is an arduous requirement when cutting hard rocks in an uncomfortable position.

A greatly improved rig is contemplated, and may include the following features: A lightweight aluminum base (of I-beam?) which by expansion-bolts in small holes can be strapped firmly to the outcrop, carrying a cutter-head mounted on universal swivels in such a way that forces are balanced so as to minimize clamping requirements, a compressed-air or gasoline-driven motor translating with the bit and in-line feed by compressed air operating on a piston. Such a rig could be quickly reduced to lightweight components and could probably be so designed that one man could operate it effectively.

The orientation of the core is accurately established by a device which slips around the core while it is still standing in the outcrop. It consists of a three-inch length of 1″ (I.D.) brass tubing, with a 1/16″ wide slot along its length. At one end of it is a flat surface, normal to the tube axis, adapted to support a standard Brunton-type geologist's pocket transit. When the proper edge of the flat plate has been leveled, the slot of the tube is running along the top element of the cylindrical core surface, and then a thin brass wire is run up and down the slot so as to leave a streak of brass on the side of the core. The inclination of the core axis and its azimuth are determined by measurements on the flat plate. The core is then broken away from its base (frequently completely, merely by sideways force at the exposed end), and the outer end is marked. The measurements of the magnetization are made relative to the brass scribe line and the mark indicating the exposed end of the core.

With this device, hundreds of cores have been oriented, but its practical design could be greatly improved. An immediate improvement would be the elimination of the requirement for the many operations with the pocket transit. Instead, the device should be equipped with level bubbles, a liquid-damped compass, and an inclinometer. Also, a system using spring members to grip the core rather than a rigid tube could be made somewhat more accurate and convenient.

When the drilling-rig cannot be used, samples are oriented by selecting a flat surface on which orientation measurements can be made. The strike is indicated by an arrow and its azimuth determined relative to magnetic north. The dip of the surface is indicated by another mark. Cores are cut from the block on the drill-press, the direction of coring being normal to the orientation plane. The orientation of the cores is established in the same manner as when they are cut *in situ*. The accuracy of this method of sampling is less than with the drill-rig, owing chiefly to the difficulties of finding a completely flat surface and then cutting cores perpendicular to it.

SPINNER MAGNETOMETER

This magnetometer is a further development of the type first pioneered by Johnson [12]. Dr. H. E. Tatel and Mr. P. F. Michelsen were responsible, in a large measure, for the success of this instrument. In it, the sample, a core one inch in diameter by an inch long, is held in a plastic (Lucite) air-driven centrifuge

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(Beams type) and spun within a pick-up coil at 282 cps. A phase reference voltage, against which the signal induced in the coil is compared, is obtained by incorporating the revolving sample holder in an optical system consisting of a line filament, a cylindrical lens, the air turbine, and a photocell. A sanded portion of the otherwise transparent plastic rotor interrupts the light. The output of the photocell passes through suitable amplifiers and filters and a variable phase shifter to the phase detector. The pick-up coil, of balanced "pancake" design, is matched by a special transformer to the input grid of the signal amplifiers. The signal is suitably amplified and passed through narrow band-pass filters to the phase detector. The photocell and signal channels are nearly identical as far as phase shift *vs* frequency characteristics are concerned, so that speed control of the rotor is not critical. Speed of rotation, determined by a discriminator circuit connected to the photocell channel, is controlled by a manual valve on the air supply. A vacuum-tube voltmeter is used to measure the voltage output of the signal channel. The gain of the signal channel can be adjusted over a range of 100 db by a variable attenuator near the beginning of the signal amplifier; another 40-db gain control is available in the unit amplifiers which follow.

The limit of sensitivity of the magnetometer, determined by extrapolation of measurements on magnets which were calibrated by the United States Coast and Geodetic Survey, is about 10^{-8} cgs/cc when working with the one-inch cylinders. This figure represents a practical limit imposed by the magnetic moment of the available plastics, and by problems of coil microphonics; it probably could be improved somewhat. The equipment has been repeatedly used in the truck for measuring directions of magnetization at this sensitivity level in such adverse locations as below power-lines and adjacent to busy highways. In some cases, it has been found necessary to null the earth's field at the sample in order to overcome the influence of a magnetically soft component which was causing discrepancies in the measurements. Normally this precaution is not required. Difficulties of measurements on inhomogeneous samples have not been recognized. Anisotropy of susceptibility of samples does not usually effect the measurement accuracy.

The direction of magnetization of the sample is determined by six measurements treated in three pairs. The angular departure of the half plane containing the N-seeking end of the magnetization direction from a reference axis that is normal to the spinning axis, is measured; the sample is then rotated 180° about the reference axis, and the angular departure of the N-seeking half plane in this new position is determined. By symmetry, the sum of the two angles must be 360°, and, if necessary, arithmetical corrections are applied to make them so. This procedure effectively calibrates the whole system for phase fidelity under operating conditions without recourse to a reference standard. Two other pairs of angles are determined in the same way by spinning the sample on other coordinate axes, and the three essential angles corresponding to the three positions are plotted as planes on an equal-area projection net [10, p. 162]. Any discrepancy in closure of the three angles on the net is recognized as a measurement error and its size is gauged by the radius of the circle that can be inscribed in the error triangle. The error radii on samples whose moments are greater than 10^{-7} cgs/cc rarely exceed one degree. The measuring and plotting require about ten minutes per sample.

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OBSERVATIONS

In Table 1, the locations where data were obtained during this special expedition to southwest United States have been indicated. A great number of excellent

TABLE 1—*Information on localities sampled*

Site No.	Locality	Latitude (north)	Longitude (west)	Age and formation	Stratigraphic thickness sampled	Lateral extent sampled	Number of oriented samples	Number of measured specimens	Literature reference
		° ′	° ′		*feet*				
				Jurassic					
1	Montoya, N.M.	35 02	104 05	Redonda Fm.	~150	1/2 mile	17	20	13
				U. Triassic					
2	Shyrocks Mill, Md.	39 33	77 23	New Oxford	20	300 ft	14	14	14
3	Las Vegas, N.M. (25 miles southeast)	35 23	104 47	Chinle	5	100 ft	6	8	15
4	Romeroville, N.M.	35 31	105 15	Chinle	15	100 ft	16	16	15
				M. Triassic					
5	Cameron, Ariz.	35 53	111 25	Shinerump	5	20 ft	17	20	16
				Permian					
6	Goulding's Trad. Post, Monument Valley, Utah	37 01	109 58	Cutler Fm., Organ Rock Tongue	200	1/4 mile	12	12	17
7	Tecolote, N.M.	35 26	105 17	Sangre de Cristo	10	100 ft	19	19	15
8	Tecolote, N.M.	35 28	105 16	Yeso	36	200 ft	26	26	15
9	Zuni Mountains, N.M.	35 21	108 26	Abo	25	150 ft	13	25	18
10	Abo Canyon, N.M.	34 27	106 23	Abo	50	400 ft	11	20	19
11	Oak Creek Canyon, Ariz.	34 56	111 41	Supai	100	1.4 miles	17	52	20
12	Carrizo Creek, Ariz.	33 57	110 18	Supai	~1,000	10 miles	30	59	20
13	Clarington, Ohio (1950 data)	39 46	80 54	Green and Washington	660	2 miles	25	32	21
14	Glenwood Spgs., Colo. (1948 data)	39 34	107 25	Cutler	1	50 ft	2	10	10, 22

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exposures of various ages and lithologies had to be ignored because of time limitations and the desire to make an extensive rather than intensive study. Many exposures, particularly of the Triassic Chinle formation, were not sampled because of their large grain size and, frequently, their lack of coherence. The Permian beds, in contrast, are generally finer grained, and contain numerous coherent beds from which good cores can be removed. In addition, during the early sampling, the Permian results appeared more interesting, and it thus seemed important to emphasize rocks of this age. The studies of Triassic rocks are being continued.

The procedure in studying an exposure usually was as follows: A day would be devoted to collecting as many samples as possible, either as blocks or with the drilling-rig; the necessary time would then be spent in completing the sample preparation, measurement, and data plotting. The outcrop would then be revisited for study with the magnetic data in hand, but no further specimens would be taken.

Information on the localities sampled is given in Table 1. According to C. B. Read [22], it has been found possible to make some broad generalizations about the genesis of these red beds in New Mexico and Arizona. The Chinle, Abo, and Sangre de Cristo formations are considered to have been derived from pre-Cambrian igneous and metamorphic rocks. They were deposited at grade on broad alluvial plains substantially at water level. Significant and extensive oxidation is believed to have resulted from fluctuations of the level of the ground water in the sediments, and hence it is inferred that much of the ferric iron was formed shortly after sedimentation. In contrast, the Yeso and Supai formations are believed to have been derived from the weathering and reworking of preexisting red sediments. The deposition was in restricted marine basins, where the precipitation of gypsum and other evaporites was favored. A reducing environment is inferred. Baker [17] has discussed the regional development of the Cutler formation.

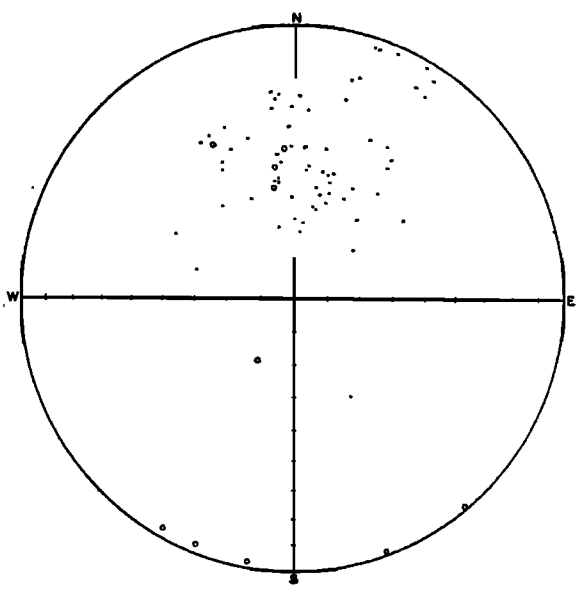


FIG. 2—Magnetizations of Triassic and Jurassic sediments

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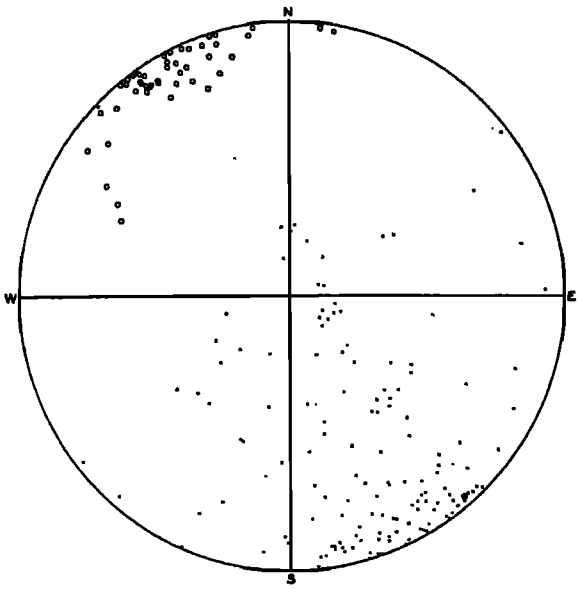


FIG. 3—Magnetizations of Permian sediments

Magnetizations observed—In Figures 2 and 3, all the directions of magnetization measured in Permian, and Triassic and Jurassic sediments, are shown in the lower hemisphere of the equal-area projection.* It is at once obvious that the two groups present a grossly different picture: In the Permian beds there is a concentration of the N-seeking direction of magnetization toward the SSE (S-seeking towards the NNW) and about horizontal; the Triassic and Jurassic beds give results not greatly different from the direction of the earth's present field, and resemble many of the results we have obtained from many different Paleozoic and Tertiary sites. There is only slight evidence of reversals of the sense of magnetization. In order to permit a rapid comparison with the results from England, the directions of the field (dipole formula—see later section) in late Paleozoic and Triassic time as reported by Clegg, *et al.*, as averages at their various sampling sites and as seen from New Mexico and Arizona, have been indicated in Figure 4. Remembering the adopted convention of plotting always in the lower hemisphere, it is seen that, although the polarizations of many samples depart greatly from Clegg's averaged directions, the majority of the Permian samples lie within the limits found in England.

THE STABILITY OF MAGNETIZATIONS

It has long been recognized that the future development of the subject of rock magnetism hinges on the question of the stability of magnetizations in time. In early work, the stability was assumed, or was inferred from simple considerations. Quantitative procedures for testing stability in special selected settings were

*This convention of plotting exclusively in the lower hemisphere differs from the convention recently adopted by some other workers of plotting the N-seeking end of the magnetization vector and indicating whether the point falls in the upper or lower hemisphere. There is obvious need for general agreement on the best procedure.

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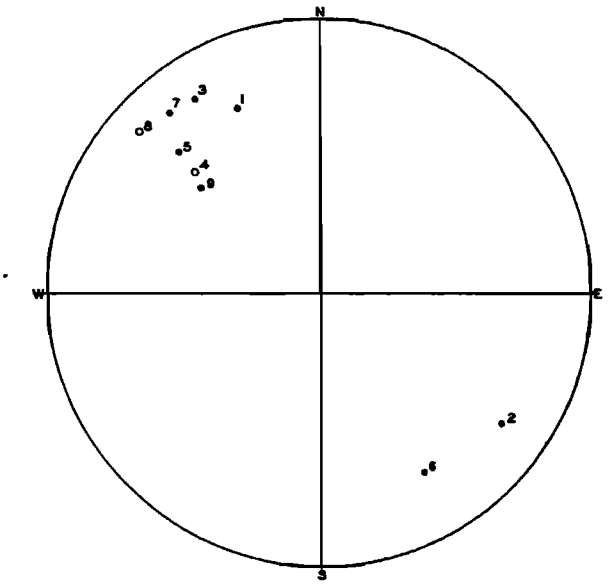


FIG. 4—Magnetizations which should be observed in Arizona for agreement with Clegg's various sites; • = N-seeking, ○ = S-seeking

formulated and first demonstrated in 1949 [10]. These procedures have found extensive use at suitable sites; they have been augmented by theoretical treatments of the properties of dispersed assemblages of tiny magnetic particles [23]. When suitable sites for stability determination are not available, indirect tests of a reassuring, though not positive nature, must be resorted to [24]. Enough observations have been made so that there is no longer any question that a useful fraction of old rocks are retaining to this day the magnetizations they received at remote times.

It seems now that still another approach to the stability question is available, and, though the work involved is tremendous, the results will be unequivocal. The approach will be illustrated by the treatment in this paper. Magnetic instability is known to result from many causes—some more effective than others—and to be certain that in a given sequence of rocks there has been *no* change of magnetization is a rigid requirement. But it seems that this requirement is met by any group of rocks that has a coherent and simple pattern of magnetization differing from the present earth's field, provided the samples come from many different places of the world. We know that many rocks can retain primary magnetizations faithfully for long times; we know also that the magnetizations of many rocks change rapidly. The geophysical problem now is to find in rocks of various ages, on a world-wide basis, the persistent and simple patterns of magnetization that have withstood all remagnetizing influences.

A simple relationship between the magnetizations of the Triassic in England and a majority of the observed magnetizations in the Permian of the southwest United States is indicated by Figure 3 and 4. This leads to the tentative conclusion that the magnetizations in question are stable. The proof of the conclusion

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will have to depend on observations made in other parts of the world. Studies in the Karoo series of Africa and its correlatives in South America and Australia should be most interesting. The relatively limited number of Triassic samples from the United States appear, for the most part, to be unstable.

THE PROBLEM OF SCATTERED POLARIZATIONS

The subject of rock magnetism has not yet developed to the point where it is possible to predict whether a given series of specimens, about which a great deal is already known geologically, will be magnetically of geophysical importance. The problem is made complicated by the great number of factors that have to do with magnetization; we are dealing with many unknowns and are groping for generalizations that will unify for us the behavior of nature.

The observations in Oak Creek Canyon, Arizona, are of interest in the problem of the scatter of magnetizations. In Figure 7 (at end of paper), sites 11*A* and 11*B*, the results from flat-lying beds and from overlying conglomerates of essentially the same material have been plotted separately. The non-random grouping indicates that the conglomerates have been magnetized following deposition, and it is thus necessary to believe that the flat beds also may have received some magnetization following the one they presumably received at the time of deposition. It is significant to note two particular features in these two plots: (1) The magnetizations found in the flat beds make a streak on the plot up into the zone where most of the conglomerate polarizations fall; and (2) the streak is roughly along a great circle between a direction close to the earth's present field and the field postulated for Permo-Triassic time. A reasonable interpretation of these observations is that initially the beds were magnetized by the field indicated by most of the other Permian samples and then later were "overprinted," in part, by new directions of magnetization in a changed earth's field. The tendency to a similar streaking in the conglomerates suggests that they, too, were magnetized, in part, in the direction of the postulated field, following their redeposition. The scatter of a couple of the points outside of the conglomerate group suggests that these samples have retained some magnetization acquired before redeposition.

A similar streaking of magnetizations is seen in the samples from Monument Valley, Figure 7, site 6, and the same interpretation of an initial magnetization plus a later overprinting in a changed earth's field seems reasonable. That the overprinting is not a simple matter of a soft magnetization produced by the earth's present field is suggested by the observation that a ± 50-gauss field applied for one minute did not alter any of the magnetizations which are in equilibrium with the present field. The very limited data available (sampling difficulties) from the Chinle southeast of Las Vegas, New Mexico (Fig. 7, site 3), also have an appearance of "streaking," but the initial and secondary fields are not known.

Some magnetization following deformation is suspected in the samples which came from an unusual group of sediments enclosed by typical M. Triassic Shinerump conglomerate (Fig. 7, site 5). Here the specimens were obtained from two settings: Exceedingly fine-grained alternating thin layers of yellow and purple material (volcanic ash?) in the flat-lying condition, and the same material in a completely disorganized jumble. The general appearance implies that the deformation took

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place when the beds were soft. Since the magnetizations fall into a far more coherent pattern than the shapes of the folds would suggest, it follows that magnetization was after deformation. Unfortunately, the magnetic susceptibility of the samples was so low (5×10^{-6} cgs/cc) that the measurement of susceptibility anisotropy could not be used as an index of deformation [25].

The concept of an initial stable magnetization plus a later overprinting does not apply to the problem of explaining the observations in the Abo formation in the Zuni Mountains (Fig. 7, site 9). Here two contrasting types of sedimentation mechanics are represented by samples from wind-laid deposits with complex cross-bedding and water-laid deposits with fine laminations having considerable horizontal extent. The color, grain size, and composition to the naked eye of the two types are identical and their magnetization patterns are indistinguishable. This material is soft and it is easy to crumble it down to individual grains. The fraction that could be extracted with a saturated Alnico V bar magnet is found to have a grain size comparable to that of the clastic quartz grains, that is, about 0.1 mm. It seems reasonable that owing to the coarse grain size of the magnetic fraction the conditions were never proper for achieving uniform magnetization at deposition. The Sangre de Cristo formation (Fig. 7, site 7) is very similar and the same situation may have applied to it. In the beds in Abo Canyon (Fig. 7, site 10), it appears that there were several layers of sufficiently fine grain to acquire a magnetization parallel to the applied field.

THE PROBLEM OF THE UNIFORM MAGNETIZATIONS

There seems to be no lack of mechanisms by which it is possible to explain away the polarizations that fail to fall in with a group. The problem of deducing the magnetization mechanism that produces a grouping of polarizations is far more subtle, and is also far more important because it involves the concept of time and dating. It is not known whether these groupings of magnetizations date mainly to the time the beds were originally deposited, or whether they date to some time after deposition. Many samples from Carrizo Creek (site 12) favor the latter possibility. Several samples have conspicuous cross-bedding, and yet their polarizations are the same in intensity and direction as those from some evenly bedded finer and coarser sediments. A couple of samples came from a very fine-grained bed, which seems to have suffered some heterogeneous distortion by being forced down into yielding underlying shale; the polarizations of this bed, however, fit in with the rest of the group.

In other rock magnetism studies, it has been demonstrated repeatedly that it is not possible to generalize about the properties of many different rocks from the properties observed in a few selected individuals. The same caution should be used in efforts to deduce the date and mechanism of magnetization of these rocks. Even though in a number of cases it is clear that magnetization followed deposition, it does not necessarily follow that this generalization applies to all samples. Until more information is available, it is not possible to say whether the magnetizations of these beds date to the time of deposition or to some later time. Many studies in younger sediments will be needed to establish the most recent date when the earth's field was in the orientation deduced from these samples.

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THE INFLUENCE OF COMPACTION

If the magnetizations observed in these strata are regarded as resulting originally by processes taking place at the time of sedimentation, then the effects of compaction must be considered. By and large, the samples from the United States are moderately fine-grained silts, in which the constituent clastic grains are in contact with one another. Material of this kind is not subject to the appreciable compaction demonstrated by finer grained sediments containing abundant clay and flocculated aggregates that incorporate quantities of water [26]. No field evidence of compaction of the beds sampled was found; therefore, it is suspected that compaction had no significant influence on primary magnetizations. Furthermore, the present magnetizations are essentially parallel to the bedding, and in this favorable orientation the effects of compaction should be minimal.

In the case of the English sediments, the polarizations are oblique to the bedding and, as pointed out by Clegg, *et al.*, compaction would act so as to decrease the original inclinations. However, the consistency of the two sets of observations implies either that compaction had no effect, or that magnetization took place after compaction.

ROTATION

One possible error that is difficult to evaluate can arise from the rotation of the outcrop about a vertical axis. This possibility is especially real in a province that has suffered from deformation, and only when the outcrops are frequent and the structures simple and known in detail can the possibility of rotation be intelligently dismissed. There is no geological basis for believing that the rocks sampled in north central New Mexico (Yeso Fm.) have been rotated 17° clockwise relative to those of central Arizona (Supai Fm.) as might be concluded from the magnetic data, but this possibility must be borne in mind.

THE PROBLEM OF THE SENSE OF THE PERMO-TRIASSIC FIELD

There are now available for consideration a great number of studies on the question of whether the sense of magnetization of a sample indicates the sense of the earth's magnetic field present when the sample was formed, or the opposite sense (for the latest discussions, see *Advances in Physics*, April 1955, for review articles by Nicholls, Néel, and Runcorn). The problem is one of mineralogy and solid state physics and chemistry, and so far has not been amenable to solution solely by field studies and geophysical considerations. Whereas recent field studies in Iceland are convincingly in favor of reversals of the earth's magnetic field [27], other observations in Japanese Pleistocene lavas are equally impressive from the standpoint of the self-reversal of the remanent moment [28]. It seems that there are subtle factors involved that have not yet been recognized; it may be that both types of reversal occur.

The data presented here on the magnetizations of the Permian sediments, when taken at their face value, imply that the north magnetic pole was formerly in the South Atlantic Ocean. This sense is indicated almost without exception by the samples from many localities and environments of deposition, and without regard to the color of the sediment, that is, whether orange, ruby red, purple, brown, pale grey-green, mottled or white. It follows, therefore, that the varied circumstances

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giving rise to the different colors of these beds (that is, different iron compounds in various states of subdivision) had no control over the sense of magnetization. Remembering this, we must attempt to visualize the behavior of the ferromagnetic minerals under the conditions governing the color of the beds. We must postulate either that the moments always undergo self-reversal, in spite of the varied environment giving rise to the different iron colors, or that they never reverse but reflect accurately the sense of the applied earth's field.

If we knew that the only magnetization events in the lives of the rocks dated to the time and conditions creating the iron colors, then we should reasonably conclude that these magnetizations reflect the sense of the applied field, for as shown by Néel [29] the conditions required for self-reversal of the moment are rather special and can be dependent on the environment.

But suppose that there were two events in the magnetization history of these rocks: first, a primary magnetization at deposition, which in the next stage dictated the sense of magnetization of secondary ferromagnetic minerals that developed under the influence of a changed environment, when, possibly, the iron colors were developed (deeper burial, higher temperatures, percolating solutions, etc.). Then we might—even though it seems unreasonable—find suitable conditions for reversing the sense of all primary magnetizations of these samples.

This argument certainly will bear no weight with those who prefer to reverse the sense of the earth's field rather than the sense of magnetization of a sample. It is advanced solely to emphasize that we do not know in detail how these rocks were magnetized or what minerals are responsible for the magnetic moment or precisely when or how the beds received their iron content, or what their thermal history has been. Even though it is easy and even reasonable to assume that these rocks reflect the sense of the old applied field, this simple assumption may not do the problem justice in the present state of ignorance of all investigators. Certainly this problem merits further study [30].

THE LOCATION OF THE MAGNETIC AXIS FROM DECLINATION AND INCLINATION MEASUREMENTS

On a sphere which either is uniformly magnetized or which contains at its center a magnetic dipole, the angle I made by the magnetic lines of force with the surface is related to the angular distance θ of the point of observation from the point where the lines of force are vertical (magnetic pole) by the expression $\tan I = 2 \cot \theta$. The magnetic field of the earth is not precisely described by this expression, for in addition to a simple main dipole component which is not exactly centered there are other more complex components. Nevertheless, this expression establishes reasonably well the great-circle distance of the magnetic pole from the point of observation. The direction to the pole is taken as the direction of the horizontal component (declination) at the point of observation. This again is an approximation when dealing with a magnetized sphere having a non-uniform magnetization. An example of the applicability of these approximations is given in Figure 5, where there are plotted the positions of the present magnetic N pole determined from several places along the equator using observed declination and inclination values and the dipole formula. The poorest determination places the

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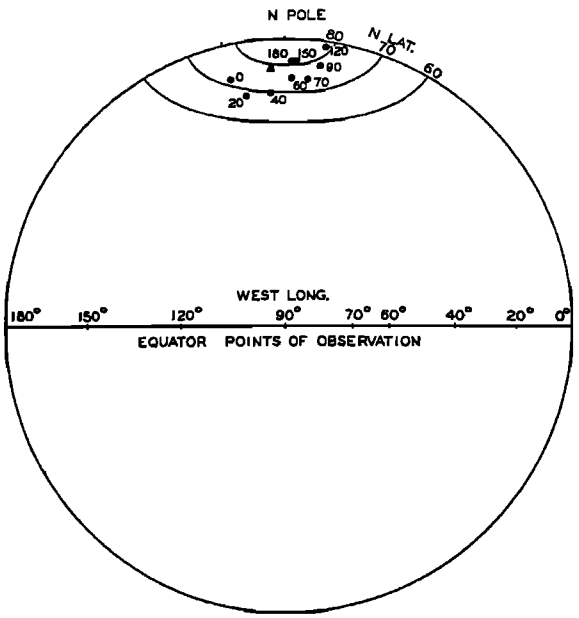


FIG. 5—Fixes on magnetic pole by dipole formula using declination and inclination values at the equator (western hemisphere, equal-area map)

pole 15° from its correct location; the average of several determinations will be close to the actual pole.

In spite of these demonstrated inadequacies, the dipole formula is the best and only one which can be used for pole determinations by rock-magnetism declination and inclination measurements; it will be possible to eliminate errors resulting from the approximation only by many observations. Inasmuch as we have no information yet on the complexity of the earth's field at remote times, we must anticipate that pole determinations made from a limited number of observations will be somewhat uncertain.

GRAPHICAL SOLUTION OF POLE LOCATION PROBLEM

The problem of locating the magnetic pole on the earth by declination and inclination measurements and the dipole formula is conveniently solved graphically by means of the equal-area (or stereographic) projection net. The net is considered to represent a global hemisphere, the small circles indicating latitude and the great circles the meridians of longitude. The site of observation is plotted at its correct latitude and longitude. The declination gives the angle that the great circle joining the magnetic pole and the site must make with the meridian through the site. This great circle is constructed as follows: A great circle 90° from the site is drawn. Along it, an angle equal to the declination is measured (in the right direction) from the meridian that passes through the site. Here a point is plotted. The magnetic pole lies on the great circle joining this point with the observation site at an angular great-circle distance from the site determined from the observed inclination and the dipole formula. The geographical coordinates of the plotted pole can then be read directly.

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POLE LOCATIONS FROM PERMO-TRIASSIC SEDIMENTS

Using the foregoing procedures, the data of Clegg, *et al.*, for nine sites in the Triassic of England have been plotted to give the pole positions indicated in Figure 6 by open circles. Notice that this Figure is a map projection, not an

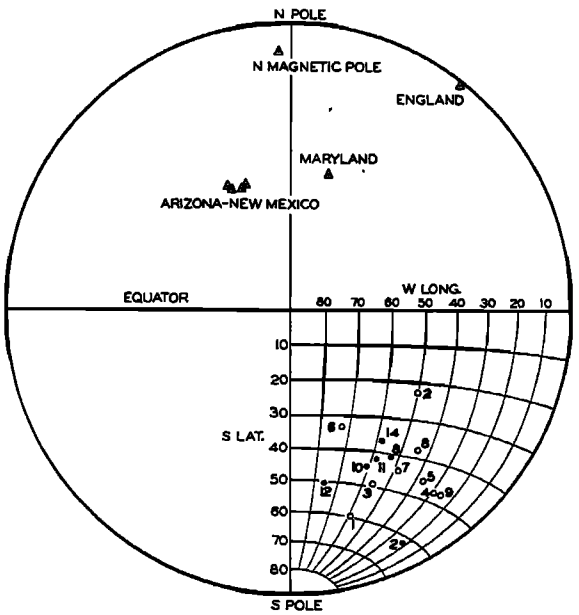


FIG. 6—Map of western hemisphere; ○ = magnetic poles located from England; ● = magnetic poles located from United States

equal-area scalar projection. The numbers correspond to those used by Clegg for his sites. The black dots indicate poles determined from Permian and Triassic rocks in the United States; the numbers refer to the sites listed in Table 1. The following procedures, numbered according to the outcrop to which they apply, were used in selecting the values of declination and inclination which give the indicated magnetic poles. All beds are assumed to have been horizontal when magnetized.

Site 2. Maryland Triassic—The average was chosen as the center of the consistent group of polarizations in fine-grained red silts; the scattered directions from coarse greywackes were ignored.

Site 12. The observed polarizations were rotated 90° on a horizontal axis (see Fig. 7, site 12*B*) in order to eliminate the projection distortion by bringing the group into the center of the net; the best average was then selected by eye. This point was then rotated back 90° to give the actual declination and inclination of the average. This procedure, though less accurate than the method provided by Fisher [31], is faster, and for the present purposes is considered sufficiently reliable.

Site 8. Yeso formation—Essentially the same procedure, the four conspicuously divergent points being ignored.

Site 11. Oak Creek Canyon—Here the selection of an average had to be

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subjective because of the streaking of the polarizations. The average was chosen as the center of the 22 points making a group centering about on the horizon.

Site 10. Abo Canyon—As in the previous case, the average was chosen as being in the center of eight points forming a group.

Site 14. Glenwood Springs, Colorado (1948 data, ref. 10, p. 148)—The necessary small corrections for the flexure fold were carried out as follows: First the fold axis was made horizontal by rotating it (and the bedding poles) on a horizontal axis at right-angles to it; the beds were then rotated on this new fold axis to make them horizontal. The polarizations were given the same rotations and their final position for flat-lying beds is shown in Figure 7, site 14. Each point represents the average of five other points which were within 4° of one another.

THE SCATTER AMONG THE DIFFERENT DETERMINATIONS

Figure 6 gives evidence that the magnetic pole in Permo-Triassic time was significantly removed from its present location. It is now important to inquire into the reasons for the lack of precise agreement among the various determinations of the pole location. Errors from experimental procedures can probably be dismissed. In those cases where large numbers of samples were individually oriented and measured, the chances of being misled by occasional mistakes (in reading the compass, for example) are minimized. The moments of all samples are sufficiently large that errors in measuring directions of magnetization are small, and errors in plotting, computation, etc., are likewise considerably smaller than the differences that appear from outcrop to outcrop.

An error still to be considered can arise from the geologists' inability to determine initial dips. When magnetizations are found parallel to the bed (horizontal field), the magnetic pole is determined as being 90° away on the sphere. If these beds had had an initial dip of 5°, the pole location would be in error by only 2.5°. In contrast, if magnetizations are inclined 70° to the bed, an uncertainty of ± 5° in the initial dip introduces an uncertainty of ± 7.5° in the location of the pole. For this reason, in addition to those cited in the discussion of compaction, it is best to determine former locations of the poles from deposits that were formed near the (then) magnetic equator. The present data fulfill these requirements well, and the discrepancies in locating the pole cannot be ascribed to uncertainties about compaction or initial dips.

It appears that the scatter among the different determinations is real; it is reasonable to assume that it represents a measure of the variability of the earth's field during the period of magnetization of these beds.

THE AGREEMENT AMONG THE DIFFERENT DETERMINATIONS

Even though the positions of the pole as determined from observations in Permo-Triassic rocks are somewhat scattered, some limits can be set on the amount that England or the United States could have rotated since these rocks were magnetized. A rotation of either land mass of 20° would cause obvious discord of the two sets of determinations. There is no way by which both England and the United States could have rotated and preserved the agreement of the observations. Likewise, a simple translation of either land mass in any direction of 20° great-circle distance would eliminate the agreement. Any amount of combined translation

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and rotation about the determined pole position as a center would preserve the agreement; this unlikely possibility can be examined by determinations in rocks of another age when the magnetic pole was in some other position.

Thus, it follows that the Wegener hypothesis of continental drift finds no support in these observations. If it should turn out that measurements from other continents confirm the pole location determined from these measurements, then the Wegener hypothesis will have ceased to be a concept worth serious consideration.

THE RELATIONSHIP OF THE MAGNETIC AND GEOGRAPHIC AXES

From two distinct lines of argument, there is justification for believing that the present-day near-coincidence of the geographic and magnetic axes is not fortuitous. Experimental evidence covering a significant span of time is found in other rock magnetism studies. The declination of Pleistocene varved clays of New England over an interval of 5,000 years shows east and west departures of about 30° from geographic north; the average of the departures centers about on true north [1]. Magnetic poles, located by a great number of magnetization observations of Tertiary rocks, scatter about the present geographic poles [9, 32, 33, 34]. These observations establish that over the past 50 million years the average of the magnetic axis positions is coincident with the present geographic axis.

Runcorn [35] has discussed the theoretical aspects of the relationships of the magnetic and geographic axes; he emphasizes that owing to Coriolis forces the most likely form that the core fluid motions would take would be in cylindrical vortices with their axes parallel to the axis of revolution of the earth. Inasmuch as modern theories on the origin of the earth's magnetic field invoke fluid motions as essential parts of the field generators, it would seem that long period symmetry of the field about the geographic axis is to be expected.

It follows, even though our understanding of the origin of the earth's magnetic field is incomplete [36], that observations over a long period of the positions of the magnetic poles will give a fix on the geographic axis.

Therefore, it is believed that, taken together, the observations reported earlier from England and those reported here provide a basis for believing that the the geographic axis in Permo-Triassic time, or possibly somewhat after, appeared at about longitude 50° west and latitude 45° south (130° east and 45° north).*

SIGNIFICANCE OF THE IMPLIED POLAR SHIFT

If the inference that there has been a shift of the geographic axis should bear the test of further observations and studies, then there is a basis for expecting that as a result important progress can be made in several other geological and geophysical problems.

The question of how the polar shift might be produced immediately directs attention to the interrelationship of the core, mantle, and crust. Inasmuch as

Footnote added in proof: Dr. S. K. Runcorn visited the Department of Terrestrial Magnetism on June 27, 1955, and informed us of his unpublished results of measurements on samples he had collected in the same general area in the summer of 1954 in Permian and Triassic sediments. Although he and the author do not necessarily agree fully on the interpretation, they do agree that the two sets of data are strikingly similar and show a highly non-statistical behavior of the earth's magnetic field during the ancient epoch when these rocks were magnetized.

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evidence has never been found suggesting that since Paleozoic time the earth has been acted on by extra-terrestrial forces of sufficient magnitude to produce a polar shift, the forces causing the shift must be looked for within the earth itself. In such an isolated system, there must be conservation of the earth's total angular momentum. This demands that when a shift of a reference point on the surface of the earth takes place, there is a compensating shift of material within the earth. This reduces the problem to one of considering how much of the outer part of the earth behaves as a coherent unit and how much of the inner part reacts so as to conserve angular momentum. Without further information, an infinity of solutions is possible.

Intimately connected with the problem of determining the relative motions is the problem of finding the driving force that produced them. From the observed shift it is seen that the force was directed obliquely to the geographic axis; to get such a biased force requires axial asymmetry in some feature of the crust, or of the core, or of the mantle between. Treatments of this problem already appear in the literature. For examples, Vestine [37] has discussed the interrelations of the variations of the geomagnetic field, fluid motions of the core, and the rate of the earth's rotation; Vening Meinesz [38] for some time has championed the rôle of convection currents in the mantle; and most recently Gold [39] has considered the problem from the standpoint of asymmetrical changes taking place mainly in the outer 100 km of the earth.

The impact of establishing a significant shift of the geographic axis on our thinking in the subjects of paleontology, stratigraphy, oceanography, climatology, etc., is obvious.

It would thus seem that important problems dealing with the earth's long-time behavior will be made more accessible to quantitative treatment when the promising possibilities of rock magnetism studies are realized.

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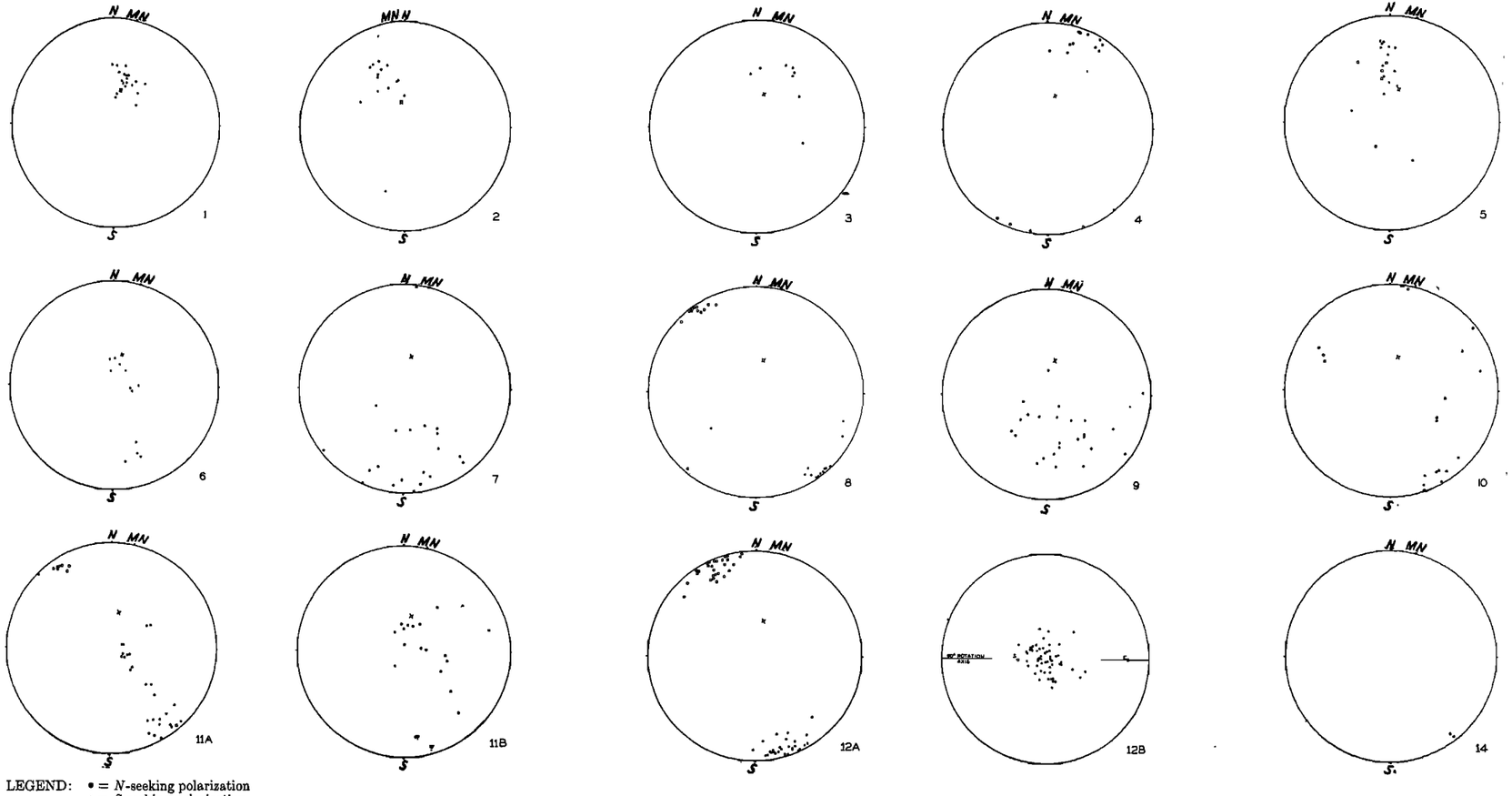


FIG. 7—Directions of magnetization obtained at sites indicated in Table 1 (p. 334)

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